Shantanu Shastri

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EDUCATION

Carnegie Mellon University December 2026
Master of Information Systems Management GPA: 3.85

Manipal Institute of Technology August 2022
Bachelor of Electronics and Communications Engineering GPA: 3.99

Relevant Coursework: Distributed Systems, Advanced Databases, Data Structures, Algorithms, System Design

SKILLS

Languages: GoLang, Java, Python, C++, TypeScript, JavaScript, SQL, Bash/Shell
Databases & Storage: Distributed Storage Systems, NoSQL (Cassandra, DynamoDB, Milvus, Neo4j), PostgreSQL
Caching & Abstraction: Redis, Memcached, Data Abstraction Services, Caching Strategies, High-Throughput Design
Orchestration & DevOps: Temporal, Kafka, Zookeeper, Docker, Kubernetes, CI/CD

EXPERIENCE

BlackRock January 2022 – August 2026
AI Engineer Internship June 2026 – August 2026

- **Data Abstraction & Orchestration:** Built a highly reliable orchestration layer and data abstraction service, unifying lifecycle management for databases and streaming systems via GoLang and Java.
- **NoSQL & Stateful Workloads:** Operated stateful workloads at scale by architecting a hybrid NoSQL distributed storage system linking Milvus embeddings to Neo4j via PostgreSQL.
- **Long-running Workflows:** Modeled long-running, stateful workflows to track data drift and latency, securely processing 3.5M records and $14T in cash with strong consistency and data reliability.

Software Developer 2 - Associate January 2025 – July 2025

- **Distributed Systems & Multi-threading:** Architected a Java Spring Boot scalable data framework, leveraging multi-threading to automate 1.5M+ client touchpoints and improve overall data reliability.
- **Distributed Caching Strategies:** Implemented distributed caching strategies using a read-through Ignite cache (similar to Redis) with LRU eviction, tailoring low-latency workloads from 450ms to 150ms.

Software Developer 1 - Analyst July 2022 – December 2024

- **High-Throughput Infrastructure:** Engineered distributed data infrastructure by sharding a server environment with Zookeeper, boosting system throughput to 10K+ RPS.
- **Database Optimization:** Optimized distributed storage systems and refactored PostgreSQL schemas on legacy tables exceeding 50M+ rows, achieving a 37% reduction in query latency.
- **Foundational Reliability:** Developed foundational infrastructure by building comprehensive Azure DevOps CI/CD pipelines with integrated testing suites for robust, highly available deployments.

Shunya June 2020 – September 2020
Data Science Intern

- **Linux Workflows:** Engineered a GPU-accelerated Python CV workflow on a Linux environment, successfully automating complex evaluations and reducing manual grading time by 90%.
- **Data Pipelines:** Addressed learning gaps by designing efficient data pipelines that analyzed multidimensional assessment data, dynamically recommending targeted practice to users.

PROJECTS

High-Performance Modular Booking Architecture

- **Distributed Storage Architecture:** Engineered a scalable storage architecture using Apache Kafka and REST APIs, seamlessly handling multi-million-QPS scale bursts with minimal latency.

Zulip Open-Source Engine Extensions

- **Stateful Data Streams:** Built a high-performance message engine using Node.js and Redis to manage stateful data streams, focusing on high reliability, clear customer impact, and zero data loss.